

# DP Computer Science: A 2.3.3 Packet Switching

---

## 1. Core Concepts & Learning Objectives

### Learning Intentions

By the end of this lesson, students will be able to:

- Explain the **packet-switching process**, including data segmentation, header information, and independent transmission
  - Describe the role of **switches** and **routers** in packet switching
  - Analyse the **advantages** of packet switching regarding **efficiency, flexibility, and fault tolerance**
  - Apply understanding of packet switching to **real-world network scenarios**
- 

### Key Vocabulary

| Term             | Definition                                                                                             |
|------------------|--------------------------------------------------------------------------------------------------------|
| Packet switching | A method of data transmission where data is split into packets and sent independently across a network |
| Packet           | A small unit of data transmitted over a network, containing a header and payload                       |
| Header           | The part of a packet containing routing information (source/destination addresses, sequence number)    |
| Payload          | The actual data being transmitted inside a packet                                                      |
| Segmentation     | The process of dividing large data into smaller packets for transmission                               |
| Reassembly       | The process of reordering packets at the destination to reconstruct the original data                  |

---

| Term            | Definition                                                                     |
|-----------------|--------------------------------------------------------------------------------|
| Switch          | A network device that forwards packets within a local area network (LAN)       |
| Router          | A network device that forwards packets between different networks              |
| MAC address     | A hardware address used by switches (unique to each network interface)         |
| IP address      | A logical address used by routers (unique identifier for devices on a network) |
| Fault tolerance | The ability of a network to continue functioning when parts fail               |

## ATL Skills

| Skill           | Description                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Thinking        | Critical thinking, analysing, understanding parts and wholes, problem-solving |
| Communication   | Explaining complex processes, active listening, note-taking                   |
| Self-Management | Time management, organisation, reflection                                     |

## 2. The Need for Packet Switching

### Why Do We Need Packet Switching?

| Challenge        | How Packet Switching Solves It                     |
|------------------|----------------------------------------------------|
| Large data files | Splits large data into smaller, manageable packets |

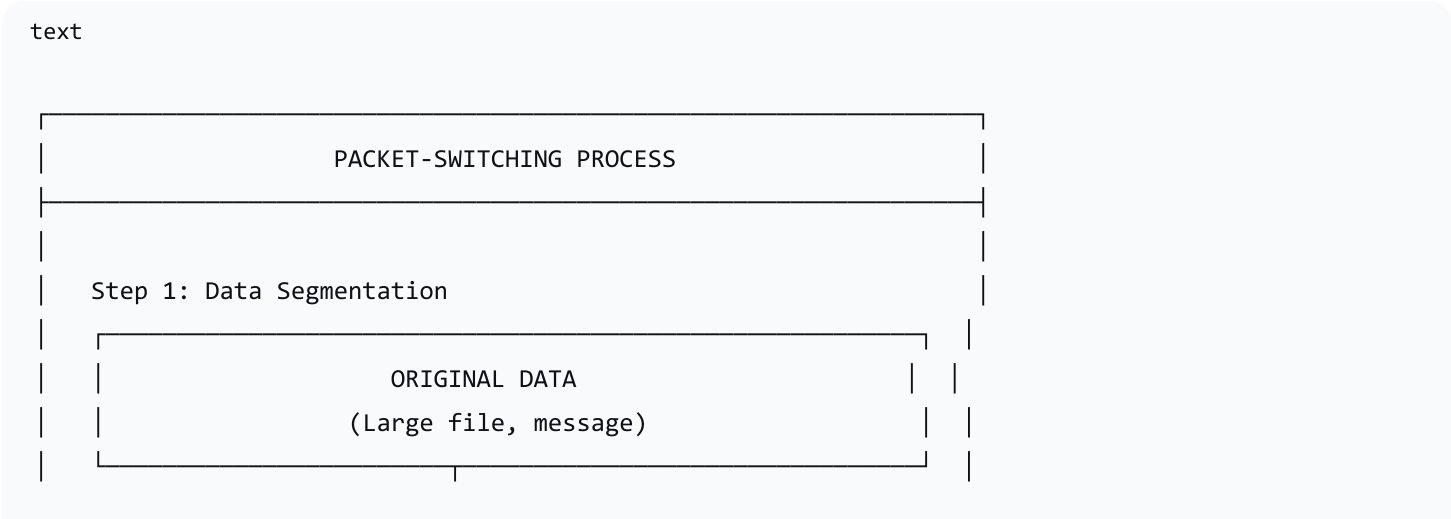
| Challenge          | How Packet Switching Solves It                               |
|--------------------|--------------------------------------------------------------|
| Network congestion | Packets can take different routes, avoiding bottlenecks      |
| Network failures   | If a route fails, packets can be re-routed                   |
| Different devices  | Packets can be delivered to any device on the network        |
| Efficiency         | Network resources are shared; no single sender blocks others |

## Analogy: Sending a Letter vs. Sending a Package

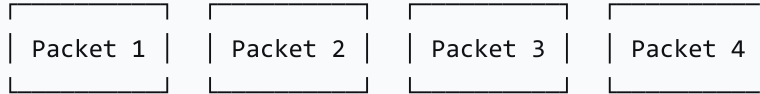
| Method            | Analogy                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Circuit Switching | Like a phone call—a dedicated connection is reserved for the entire duration                                                                           |
| Packet Switching  | Like sending a 1000-page book in 100 separate envelopes, each sent independently, potentially via different routes, and reassembled at the destination |

## 3. The Packet-Switching Process

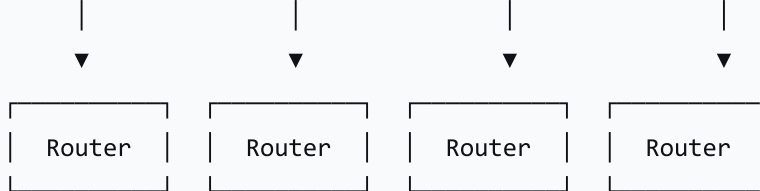
### Step-by-Step Process



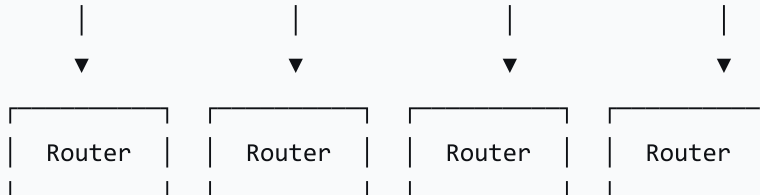
Step 2: Splitting into Packets



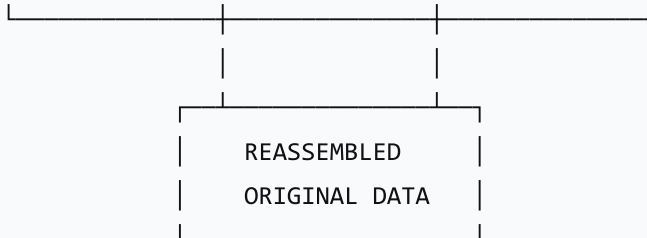
Step 3: Independent Transmission



Step 4: Different Routes



Step 5: Reassembly (at destination)

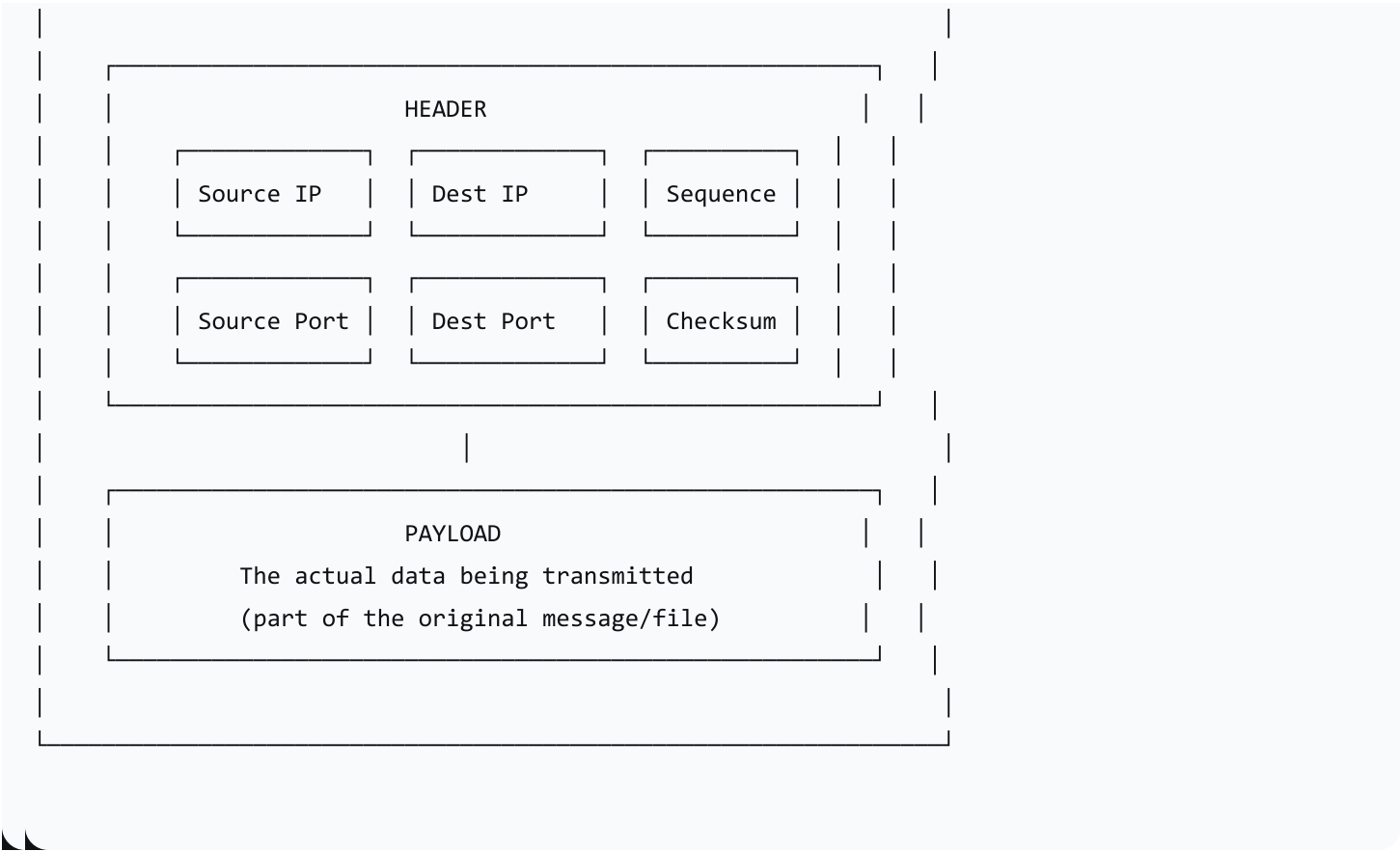


## 4. Packet Structure

### What is Inside a Packet?

text

PACKET STRUCTURE



## Header Information

| Field                  | Description             | Purpose                                    |
|------------------------|-------------------------|--------------------------------------------|
| Source IP Address      | IP address of sender    | Identifies where the packet came from      |
| Destination IP Address | IP address of receiver  | Identifies where the packet is going       |
| Source Port            | Port number of sender   | Identifies the sending application         |
| Destination Port       | Port number of receiver | Identifies the receiving application       |
| Sequence Number        | Packet order number     | Allows reassembly at destination           |
| Checksum               | Error-checking value    | Detects if packet was corrupted in transit |
| Time to Live (TTL)     | Hop limit               | Prevents packets from circulating forever  |

## 5. Switches and Routers

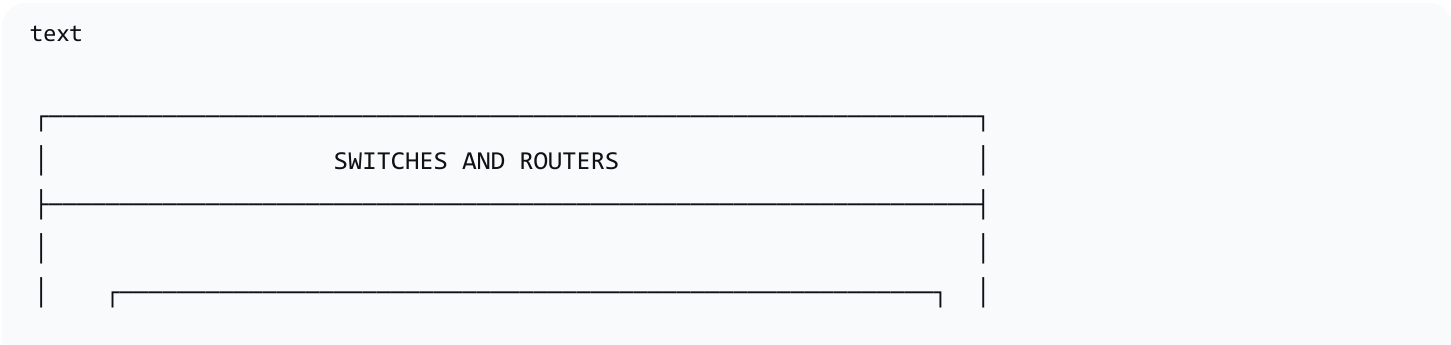
### Switches

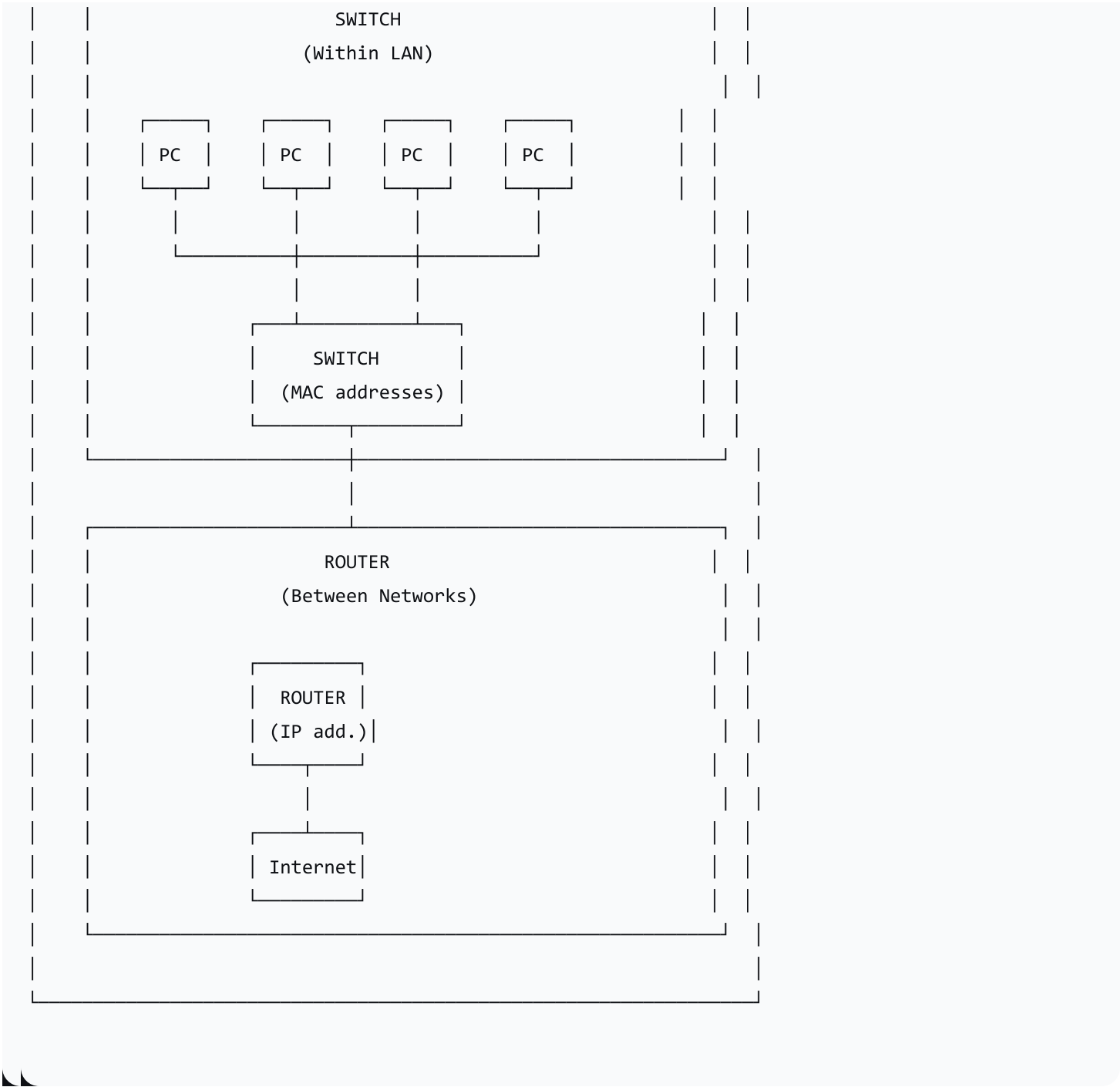
| Aspect       | Description                                                        |
|--------------|--------------------------------------------------------------------|
| Function     | Forwards packets within a <b>local area network (LAN)</b>          |
| How It Works | Uses <b>MAC addresses</b> to forward packets to the correct device |
| Layer        | Data Link Layer (Layer 2)                                          |
| Analogy      | Like a <b>mailroom</b> —delivers mail within the same building     |

### Routers

| Aspect       | Description                                                                   |
|--------------|-------------------------------------------------------------------------------|
| Function     | Forwards packets between <b>different networks</b> (e.g., LAN to internet)    |
| How It Works | Uses <b>IP addresses</b> to route packets across networks                     |
| Layer        | Network Layer (Layer 3)                                                       |
| Analogy      | Like a <b>post office</b> —sorts and sends mail to different cities/countries |

### Visual Comparison





## Switch vs. Router Comparison

| Feature      | Switch                 | Router                     |
|--------------|------------------------|----------------------------|
| Network Type | LAN (local)            | Inter-network (LAN to WAN) |
| Address Used | MAC Address (hardware) | IP Address (logical)       |
| Layer        | Layer 2 (Data Link)    | Layer 3 (Network)          |

| Feature  | Switch                       | Router                            |
|----------|------------------------------|-----------------------------------|
| Analogy  | Mailroom                     | Post Office                       |
| Function | Delivers within same network | Routes between different networks |

## 6. Advantages of Packet Switching

### Efficiency

| Aspect             | Description                                            |
|--------------------|--------------------------------------------------------|
| Shared Resources   | Network bandwidth is shared among all users            |
| No Dedicated Lines | No need to reserve a circuit for each connection       |
| Dynamic Routing    | Packets can take the best available path at any moment |
| Reduced Congestion | Traffic is spread across multiple paths                |

**Example:** During a live video stream, packets can take different routes to avoid congestion, ensuring smooth playback.

### Flexibility

| Aspect                 | Description                                               |
|------------------------|-----------------------------------------------------------|
| Multiple Data Types    | Handles voice, video, text, and files on the same network |
| Varying Packet Sizes   | Can transmit packets of different sizes                   |
| Adaptive Routing       | Can adapt to changing network conditions                  |
| Different Applications | Supports web browsing, streaming, gaming, and VoIP        |



**Example:** A network can simultaneously handle a video call (real-time) and a file download (non-critical) using the same infrastructure.

---

## Fault Tolerance

| Aspect                     | Description                                            |
|----------------------------|--------------------------------------------------------|
| No Single Point of Failure | If one router fails, packets can be re-routed          |
| Alternative Paths          | Multiple routes ensure data reaches its destination    |
| Redundancy                 | Network can survive component failures                 |
| Automatic Recovery         | Routers dynamically find new paths when failures occur |

**Example:** If a cable is cut between two cities, routers will automatically find an alternative route for packets.

---

## Comparison: Circuit Switching vs. Packet Switching

| Feature         | Circuit Switching                       | Packet Switching                    |
|-----------------|-----------------------------------------|-------------------------------------|
| Dedicated Path  | Yes (reserved for entire communication) | No (packets take independent paths) |
| Resource Use    | Fixed (wasted when idle)                | Dynamic (shared with others)        |
| Fault Tolerance | Low (if path fails, connection is lost) | High (can reroute around failures)  |
| Efficiency      | Low (resources are reserved)            | High (resources are shared)         |
| Setup Time      | High (needs connection setup)           | Low (no setup required)             |
| Example         | Traditional phone calls                 | Internet data, VoIP, streaming      |

---

## 7. Real-World Applications of Packet Switching

| Application          | Why Packet Switching?                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------|
| Video Streaming      | Splits video into packets; packets can take different routes to avoid congestion              |
| Online Gaming        | Packets sent independently; if one is lost, only that packet needs to be resent               |
| VoIP (Voice over IP) | Voice is digitised and sent in packets; efficient use of network resources                    |
| Web Browsing         | Web pages split into packets; pages load efficiently even with network congestion             |
| File Transfer        | Large files split into packets; if one packet is corrupted, only that packet is retransmitted |
| Email                | Emails are sent as packets; efficient delivery across networks                                |

## 8. Activity: Packet Switching Worksheet

### Example Questions

1. Explain the packet-switching process in your own words.
2. What information does a packet header contain? Why is this information needed?
3. Explain the difference between a switch and a router. What is each one responsible for?
4. List three advantages of packet switching and explain why each is important.
5. Scenario: You are streaming a video on your phone. How does packet switching ensure smooth playback even if the network is busy?
6. Diagram: Draw a diagram showing how a file is split into four packets, sent via different routes, and reassembled at the destination.

### Example Answers

1. Packet-Switching Process:

Data is split into smaller packets. Each packet is given a header with source/destination IP addresses and a sequence number. Packets are sent independently across the network, potentially taking different routes. At the destination, packets are reassembled in the correct order.

2. Header Information:

Source IP, destination IP, sequence number, checksum, TTL. This information is needed to route packets correctly, ensure they arrive in order, and check for errors.

3. Switch vs. Router:

- **Switch:** Forwards packets within a LAN using MAC addresses.
- **Router:** Forwards packets between different networks using IP addresses.

4. Advantages of Packet Switching:

- **Efficiency:** Network resources are shared; no dedicated lines required.
- **Flexibility:** Handles different data types and adapts to network conditions.
- **Fault Tolerance:** If a route fails, packets can be re-routed.

---

## 9. Lesson Sequence

### Lesson Plan

| Phase                | Time   | Activity                                                                                 |
|----------------------|--------|------------------------------------------------------------------------------------------|
| Introduction         | 5 min  | "How does a message get from your phone to a friend's phone?"; state learning objectives |
| Packet Dissection    | 15 min | Explain data segmentation, packet structure (header/payload); use analogies              |
| Switches and Routers | 15 min | Explain roles of switches and routers; MAC vs. IP addresses                              |
| Advantages           | 15 min | Efficiency, flexibility, fault tolerance; real-world examples                            |

| Phase           | Time   | Activity                                              |
|-----------------|--------|-------------------------------------------------------|
| Worksheet & Q&A | 10 min | Complete worksheet; review answers; address questions |

## 10. Assessment Activities

| Assessment Type     | Description                                                     |
|---------------------|-----------------------------------------------------------------|
| Class Participation | Observe engagement and contributions                            |
| Worksheet           | Assess understanding of packet-switching process and advantages |
| Quiz                | Multiple-choice quiz on eBook page A2.3.3                       |
| Exit Ticket         | Gauge understanding and identify areas needing clarification    |

### Sample Exit Ticket Questions:

1. What is packet switching?
2. What is the difference between a packet header and the payload?
3. What is the role of a router in packet switching?
4. Name one advantage of packet switching.

## 11. Differentiation

| Level     | Strategy                                                                                       |
|-----------|------------------------------------------------------------------------------------------------|
| Support   | Provide visual aids, concise definitions, and real-world examples                              |
| Challenge | Encourage research on quality of service (QoS) and how packet prioritisation works in networks |

## 12. Resources

- **Presentation** (Slide Deck)
  - **eBook** (A2.3.3 with quiz)
  - **Worksheet:** Packet-switching process + questions
  - **Worksheet Answers**
  - **Quizlet** (A2.3.3 vocabulary flashcards)
- 

## 13. Summary: Key Takeaways

| Concept          | Key Point                                           |
|------------------|-----------------------------------------------------|
| Packet Switching | Data is split into packets and sent independently   |
| Packet Structure | Header (routing info) + Payload (actual data)       |
| Switches         | Forward packets within LAN using MAC addresses      |
| Routers          | Forward packets between networks using IP addresses |
| Efficiency       | Network resources are shared; no dedicated lines    |
| Flexibility      | Handles different data types; adapts to conditions  |
| Fault Tolerance  | If a route fails, packets can be re-routed          |

"Packet switching is the foundation of modern internet communication—it is efficient, flexible, and resilient, allowing data to travel across networks reliably."